The Chinese University of Hong Kong, Shenzhen
School of Data Science

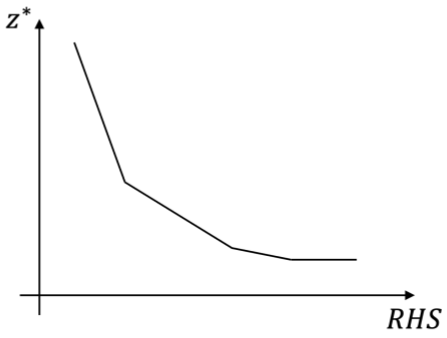
MAT3007 · Homework 5 · Spring 2024

Due: Wednesday, April 10 at 12:00pm (noon)

Please note: You should submit your answers (scanned or typeset) in a PDF fle. Question 2 and 3 asks you to write codes in Python or MATLAB. Submit all necessary code fles via Blackboard. Also, give meaningful names to your decision variables and constraints, and add comments to your code liberally. The problem set is due at 12:00pm (noon) on Wednesday, April 10th.

Problem 1

1. (20 pts) The following graph depicts the optimal value of a linear program, denoted by z^*, as the right-hand side of one of the constraints, denoted by RHS, changes.



   (a) Determine whether this LP is a maximization or minimization problem, and whether the constraintis a "$\geq$" or a "$\leq$" constraint. Explain your reasons.

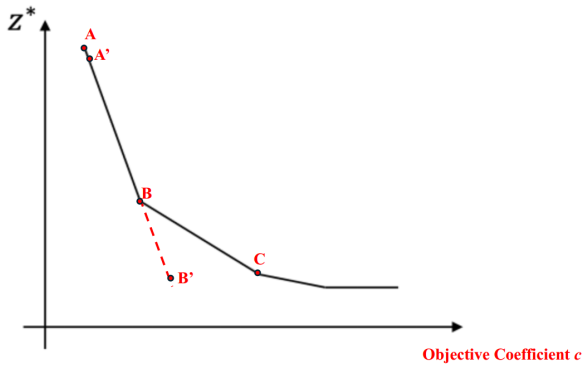
   (b) Now replace "RHS" by "objective coefficient" and interpret this graph depicting how the optimal objective changes as the objective coefficient of a primal variable is changed. Determine if the linear program is a maximization or minimization problem, and whether the (optimal) value of the primal variable is nonnegative or nonpositive.

Solution:

(a) **Minimization problem:** We consider the dual problem of this primal problem. Then the RHS in primal problem corresponds to objective coefficient in the dual problem. Because it is piecewise linear convex, the dual problem corresponds to $\max_\pi \pi^\top b$. Based on the duality theory, the primal problem is a minimization problem.
"$\leq$": We can find the constraint is "tight" until optimal value z^* is unchanged. When RHS increases, the constraint gets loose and this minimization problem get a smaller objective value, so it must be "$\leq$".

(b) **Nonpositive optimal primal variable:** When we replace "RHS" by "objective coefficient", the optimal objective z^* reduces until unchanged as the objective coefficient c of a primal

variable is increasing. Based on the local sensitivity theory, the optimal solution x^* is unchanged with a very limited change of objective coefficient c. Since the x^* is unchanged, z^* becomes smaller (Point A to Point A') when c increases, it implies the primal variable is nonpositive.



Maximization problem: We can see the straight line between point A and point B, it implies the optimal solution x^* is unchanged as the c increasing. If the primal problem is a minimization problem, then in order to obtain the optimal objective value, it will keep moving from point B to point B' instead of point C. Hence, the primal problem is a maximization problem, the optimal solution x^* in the segment AB is different from the optimal solution x^* in the segment BC. In other words, the optimal solution is changed to obtain the optimal objective value for this maximization problem.
We can also consider the dual problem of this primal problem. Based on the LP duality theory, The increasing objective coefficient c in the primal problem can be transferred to the increasing RHS in the dual problem. As we have known in (a), the increasing RHS with decreasing z^* implies the dual problem is a minimization problem with $\leq$ constraints. So we know the primal problem is a maximization problem with nonpositive variables.

Problem 2

2. (30 pts) Consider the unconstrained optimization problem

$$\min_{x\in\mathbb{R}^2} \quad f(x) := x_1^4 + \frac{2}{3}x_1^3 + \frac{1}{2}x_1^2 - 2x_1^2x_2 + \frac{4}{3}x_2^2.$$

(a) Calculate all stationary points of f and investigate whether the stationary points are local maximizer, local minimizer, or saddle points.

(b) Create a surface and/or contour plot of the function using MATLAB or Python and decide whether the problem possesses a global solution or not.

Solution :

a) We first calculate the gradient and Hessian of f:

$$\nabla f(x) = \begin{pmatrix} 4x_1^3 + 2x_1^2 + x_1 - 4x_1x_2 \\ -2x_1^2 + \frac{8}{3}x_2 \end{pmatrix}, \quad \nabla^2 f(x) = \begin{pmatrix} 12x_1^2 + 4x_1 + 1 - 4x_2 & -4x_1 \\ -4x_1 & \frac{8}{3} \end{pmatrix}.$$

We have $\nabla f(x) = 0$ if and only if $4x_2 = 3x_1^2$ and $x_1(x_1^2 + 2x_1 + 1) = 0$. This yields the two stationary points $x^* = (0, 0)$ and $y^* = (-1, \frac{3}{4})$. It holds that

$$\nabla^2 f(x^*) = \begin{pmatrix} 1 & 0 \\ 0 & \frac{8}{3} \end{pmatrix} \quad \text{and} \quad \nabla^2 f(y^*) = \begin{pmatrix} 6 & 4 \\ 4 & \frac{8}{3} \end{pmatrix}.$$

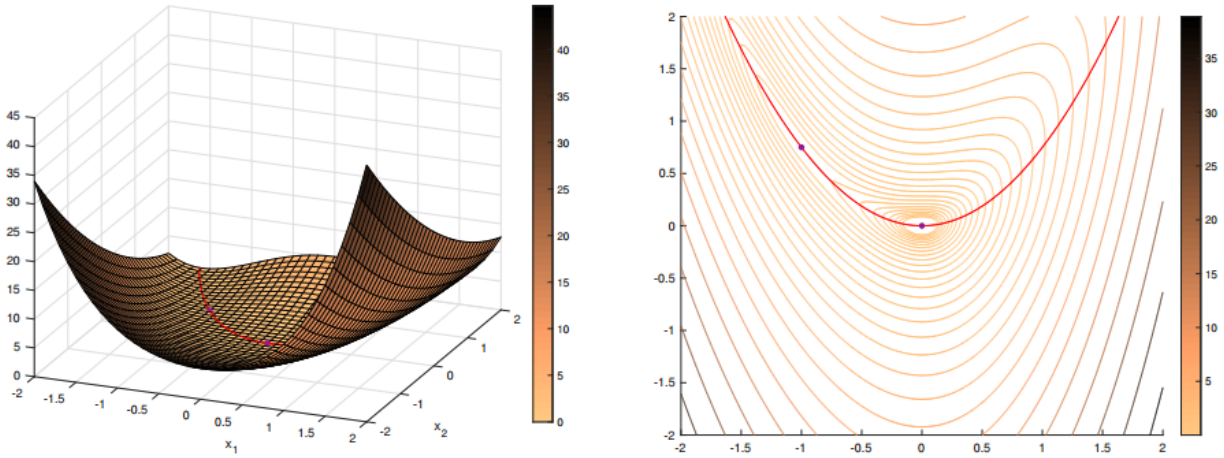
The Hessian $\nabla^2 f(x^*)$ is a diagonal matrix with eigenvalues 1 and $\frac{8}{3}$. Hence, $\nabla^2 f(x^*)$ is positive definite and x^* is a strict local minimizer of f. We further have $\det(\nabla^2 f(y^*)) = 2 \cdot 8 - 16 = 0$ and $\operatorname{tr}(\nabla^2 f(y^*)) = 6 + \frac{8}{3} > 0$. Hence, $\nabla^2 f(y^*)$ is positive semidefinite. In order to decide whether y^* is a saddle point or minimizer, we discuss the objective function around y^* in more detail. It holds that

$$\begin{aligned} f(x) &= x_1^4 + \frac{2}{3}x_1^3 + \frac{1}{2}x_1^2 - 2x_1^2x_2 + \frac{4}{3}x_2^2 = x_1^4 + \frac{2}{3}x_1^3 + \frac{1}{2}x_1^2 - \frac{3}{4}x_1^4 + \left(\frac{\sqrt{3}}{2}x_1^2 - \frac{2}{\sqrt{3}}x_2\right)^2 \\ &= \frac{x_1^2}{2}\left(\frac{1}{2}x_1^2 + \frac{4}{3}x_1 + 1\right) + \left(\frac{\sqrt{3}}{2}x_1^2 - \frac{2}{\sqrt{3}}x_2\right)^2 \end{aligned}$$

and $f(y^*) = \frac{1}{2}(\frac{1}{2} - \frac{4}{3} + 1) + 0 = \frac{1}{12}$. Our idea is to discuss the behavior of $\frac{x_1^2}{2}\left(\frac{1}{2}x_1^2 + \frac{4}{3}x_1 + 1\right)$ in a neighborhood of -1. Let us consider the path $t \mapsto \phi(t) := (t - 1, \frac{3}{4}(t-1)^2)$. Then, we have

$$f(\phi(t)) = \frac{(t-1)^2}{2}\left(\frac{1}{2}t^2 - t + \frac{1}{2} + \frac{4}{3}t - \frac{4}{3} + 1\right) = \cdots = \frac{t^4}{4} - \frac{t^3}{3} + \frac{1}{12} = f(y^*) + t^3\left(\frac{t}{4} - \frac{1}{3}\right).$$

Thus, for all $t \in (0, \frac{4}{3})$, we obtain $f(\phi(t)) < f(y^*)$ and for all $t < 0$, it holds that $f(\phi(t)) > f(y^*)$. We conclude that y^* is a saddle point of f.



(a) Surface Plot of f (b) Contour Plot of f

Figure 1: Illustration of f in A6.1

b) A surface and contour plot of f is shown in Figure 1. The red line in the plots corresponds to the function $t \mapsto \phi(t)$. The code generating these plots can be found in Listing 1 & 2.

The plots suggest that f has a global solution and that it is given by x^*.

In fact, we can also prove this mathematically. (This is not required as part of your reply). The function $g(x_1) = \frac{1}{2}x_1^2 + \frac{4}{3}x_1 + 1$ attains its global minimum at $x_1 = -\frac{4}{3}$, i.e., we have $g(x_1) \geq g(-\frac{4}{3}) = \frac{1}{9}$ for all x_1. Reusing the calculations from part a), this shows:

$$f(x) = \frac{x_1^2}{2} g(x_1) + \left(\frac{\sqrt{3}}{2} x_1^2 - \frac{2}{\sqrt{3}} x_2 \right)^2 \geq \frac{x_1^2}{18} \geq 0$$

for all $x \in \mathbb{R}^2$. Since $x^* = (0, 0)$ is a stationary point with $f(x^*) = 0$, this implies that x^* is a global minimum of f.

Problem 3

3. (30 pts) For the following function

$$f(x) = x_1 e^{-\frac{1}{2}(x_1^2+x_2^2)}.$$

(a) Write down the gradient and the Hessian matrix of $f(x)$.

(b) Derive the second-order Taylor expansion of $f(x)$ at the point $\bar{x} = [-1, \frac{1}{2}]$.

(c) Plot the function in part (a) in the area $x_1 \in [-3, 3]$ and $x_2 \in [-3, 3]$

(d) Plot the second-order Taylor expansion in part (b) in the same figure in the area $x_1 \in [-1.5, -0.5]$ and $x_2 \in [-1, 1]$. The goal is to demonstrate that the Taylor expansion approximates $f(x)$ very well close to $\bar{x}$, so please print the figure from an angle that helps to see how the function and the Taylor expansions compare.

Solution:

(a) Given gradient $\nabla f(x)$ and Hessian matrix $\nabla^2 f(x)$

$$\nabla f(x) = \begin{bmatrix} 1 - x_1^2 \\ -x_1 x_2 \end{bmatrix} * e^{-\frac{1}{2}(x_1^2+x_2^2)}.$$

$$\nabla^2 f(x) = \begin{bmatrix} x_1^3 - 3x_1 & x_2(x_1^2 - 1) \\ x_2(x_1^2 - 1) & x_1(x_2^2 - 1) \end{bmatrix} * e^{-\frac{1}{2}(x_1^2+x_2^2)}.$$

(b) The second-order Taylor expansion of $f(x)$ at point $\overline{x} = [-1, \frac{1}{2}]$ is

$$f(x) \approx f(\overline{x}) + \nabla f(\overline{x})^T (x - \overline{x}) + \frac{1}{2}(x - \overline{x})^T \nabla^2 f(\overline{x})(x - \overline{x}).$$

Here we have

$$f(\overline{x}) = -e^{-\frac{5}{8}}.$$

$$\nabla f(\overline{x}) = \begin{bmatrix} 0 \\ \frac{1}{2} \end{bmatrix} * e^{-\frac{5}{8}}.$$

$$\nabla^2 f(\overline{x}) = \begin{bmatrix} 2 & 0 \\ 0 & \frac{3}{4} \end{bmatrix} * e^{-\frac{5}{8}}.$$

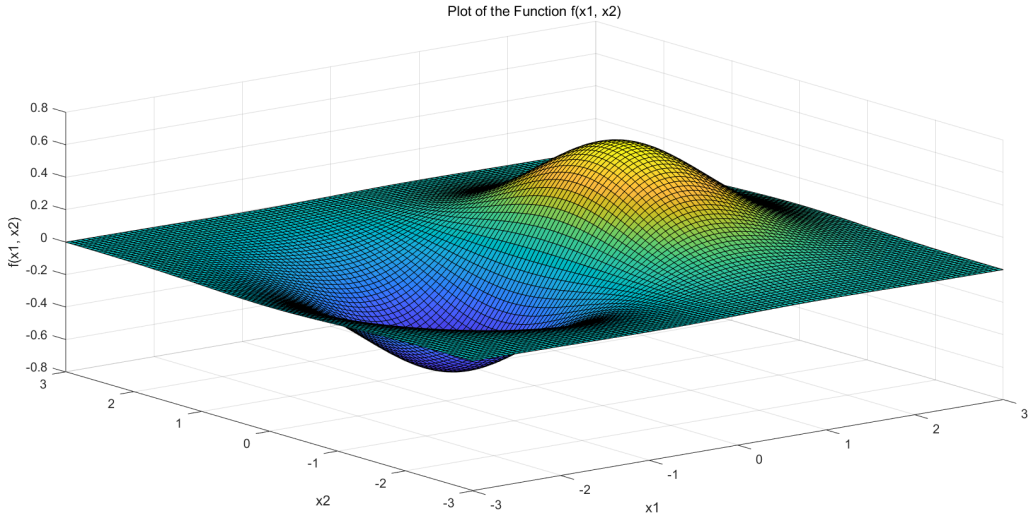
$$x - \overline{x} = \begin{bmatrix} x_1 + 1 \\ x_2 - \frac{1}{2} \end{bmatrix}.$$

Finally we will get

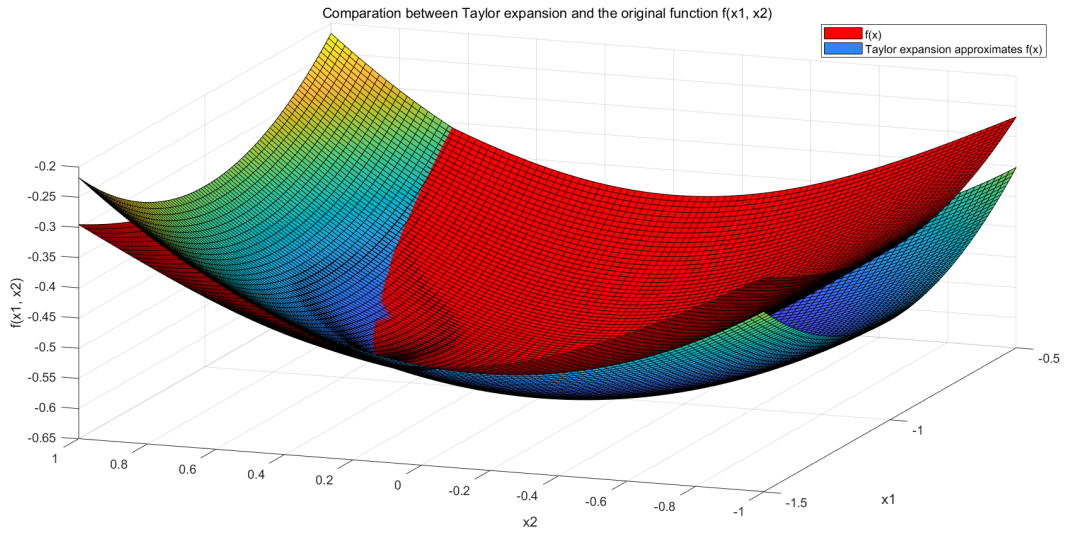
$$f(x) \approx f(\overline{x}) + \nabla f(\overline{x})^T (x - \overline{x}) + \frac{1}{2}(x - \overline{x})^T \nabla^2 f(\overline{x})(x - \overline{x})$$

$$\approx -e^{-\frac{5}{8}} + \frac{1}{2} e^{-\frac{5}{8}}(x_2 - \frac{1}{2}) + (x_1 + 1)^2 * e^{-\frac{5}{8}} + \frac{3}{8}(x_2 - \frac{1}{2})^2 * e^{-\frac{5}{8}}.$$

$$\approx e^{-\frac{5}{8}} * (x_1^2 + \frac{3}{8} x_2^2 + 2x_1 + \frac{1}{8} x_2 - \frac{5}{32}).$$


Plot of the Function f(x1, x2)
f(x1, x2)
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
3
2
1
0
-1
-2
-3
x2
-3
-2
-1
0
1
2
3
x1

(c)


Comparation between Taylor expansion and the original function f(x1, x2)
f(x)
Taylor expansion approximates f(x)
f(x1, x2)
-0.2
-0.25
-0.3
-0.35
-0.4
-0.45
-0.5
-0.55
-0.6
-0.65
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
x2
-0.5
-1
-1.5
x1

(d)

Problem 4

4. (20 pts) Consider the nonlinear optimization problem:

$$\begin{aligned} \min_{x_1,x_2,x_3} \quad & 2x_1x_2 + \frac{1}{2}x_3^2 \\ \text{s.t.} \quad & 2x_1x_3 + \frac{1}{2}x_2^2 \leq 0 \\ & 2x_2x_3 + \frac{1}{2}x_1^2 \leq 0. \end{aligned}$$

(a) Write down the KKT conditions for this problem.

(b) Investigate whether the point $x^* = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ is a KKT point.

Solution : We define $f(x) := 2x_1x_2 + \frac{1}{2}x_3^2$, $g_1(x) := 2x_1x_3 + \frac{1}{2}x_2^2$ and $g_2(x) = 2x_2x_3 + \frac{1}{2}x_1^2$. Then, it holds that

$$\nabla f(x) = \begin{pmatrix} 2x_2 \\ 2x_1 \\ x_3 \end{pmatrix}, \quad \nabla g_1(x) = \begin{pmatrix} 2x_3 \\ x_2 \\ 2x_1 \end{pmatrix}, \quad \nabla g_2(x) = \begin{pmatrix} x_1 \\ 2x_3 \\ 2x_2 \end{pmatrix}.$$

a) The KKT conditions are given by: find $x \in \mathbb{R}^3$, $\lambda \in \mathbb{R}^2$ such that

$$\begin{aligned} 2x_2 + 2x_3\lambda_1 + x_1\lambda_2 &= 0, \\ 2x_1 + x_2\lambda_1 + 2x_3\lambda_2 &= 0, \\ x_3 + 2x_1\lambda_1 + 2x_2\lambda_2 &= 0, \\ \lambda_1(2x_1x_3 + \tfrac{1}{2}x_2^2) &= 0, \\ \lambda_2(2x_2x_3 + \tfrac{1}{2}x_1^2) &= 0, \\ \lambda_1, \lambda_2 \geq 0, \quad g_1(x) \leq 0, \quad & g_2(x) \leq 0. \end{aligned}$$

And the primal conditions are as same as the constraints in the primal problem.

b) The point $x^* = (0,0,0)^\top$ obviously satisfies the KKT conditions for all $\lambda_1, \lambda_2 \geq 0$.